Korea Institute of Science and Technology

Surface analysis:

Photoelectron spectroscopy

5. Spectrum interpretation and binding energy

Initial and Final state effect

Dr. Soohyung Park

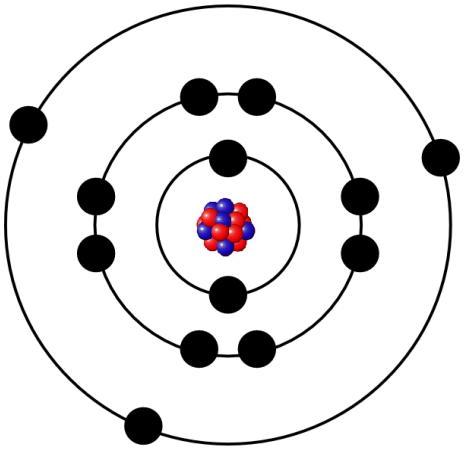
XPS spectrum interpretation

Raw Data

Information

Electronic structure

- **Atom**



- Nucleus: Protons + Neutrons
- The number of proton

  → **Atomic number Z**
- Electrons are binding at the protons by electrostatic force

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$ Coulomb's Law

Electronic structure

❖ Schrodinger Equation of Electrons under Coulomb Potential

$$H\psi(\mathbf{r}) \equiv \left[-\frac{\hbar^2}{2m_e}\nabla^2 + V(r)\right]\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

- **Solutions to the Schrödinger Equation:**

  Energy eigenstates that have allowable energy levels within given conditions, through the electron's wave function, we can perfectly describe the energetic and spatial distribution of the electron.

$$\psi_{nlm} = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}}\, e^{-r/na} \left(\frac{2r}{na}\right)^l \left[L_{n-l-1}^{2l+1}(2r/na)\right] Y_l^m(\theta,\phi)$$

Electronic structure

❖ **Quantum Numbers** (n,l,m of ψ_{nlm})

- Electrons are assigned quantum numbers n, l, and m, which determine the electron's state.

① **Principle Quantum Number: *n***

"***n*** influences both the energy and spatial distribution of electrons. It can only be a positive integer. As n increases, the binding energy (B.E) tends to decrease.

"Shell"

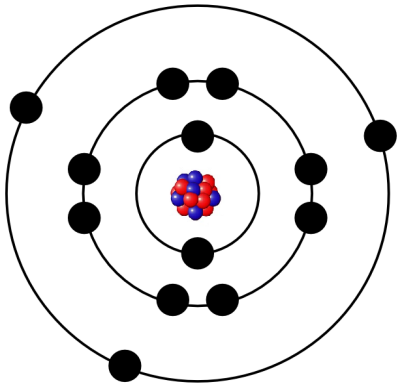
② **Angular Momentum Quantum Number: *l***

"It influences the **spatial distribution** of electrons. ***n*** must be determined first, and it can have values of l = 0, 1, … n-1. This is what we typically learned as the 'orbital'.

③ **Magnetic Quantum Number: *m*** or ***m*$_l$**

"It influences the direction of spatially distributed electrons (Direction)

l must be determined first, and it can have values of ml = -l, -l+1, …, 0, …, l-1, l."



Electronic structure notion

Spectroscopic Notation

n: number

l: for l = 0, **s orbital (Sharp)**

l = 1, **p orbital (Principle)**

l = 2, **d orbital (Diffuse)**

l = 3, **f orbital (Fundamental)**

$$n[Orbital]_j$$

"j represents the total angular momentum determined by l and s.

ex. 1)

n=2, l=0, m=0, s=+1/2 electron

$$2S_{1/2}$$

ex. 2)

n=3, l=2, m=+2, s=+1/2 electron

$$3d_{5/2}$$

In the case of the S orbital, sometimes ½ is omitted."

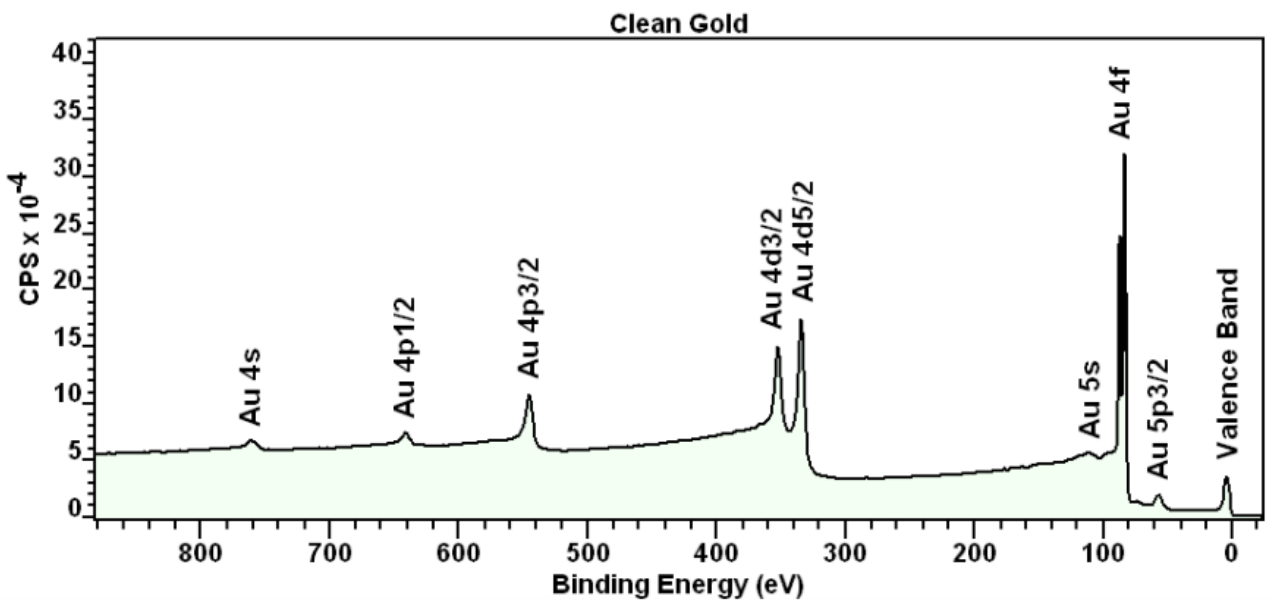
Electronic structure notion

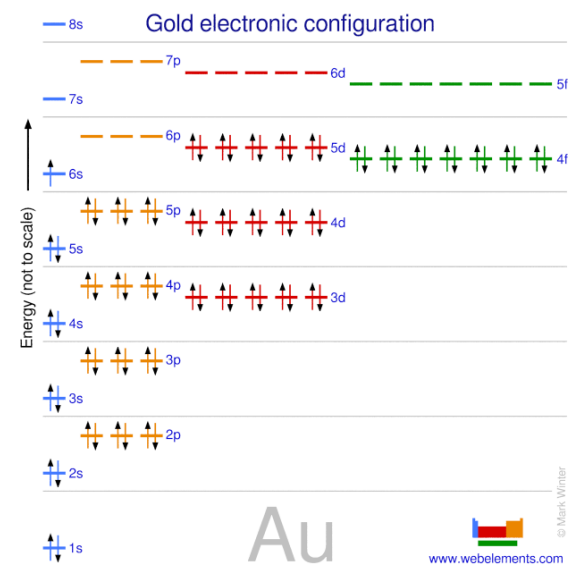
Spectroscopic Notation

$$n[Orbital]_j$$

ex. 2)

n=4, l=2, m=+2, s=+1/2

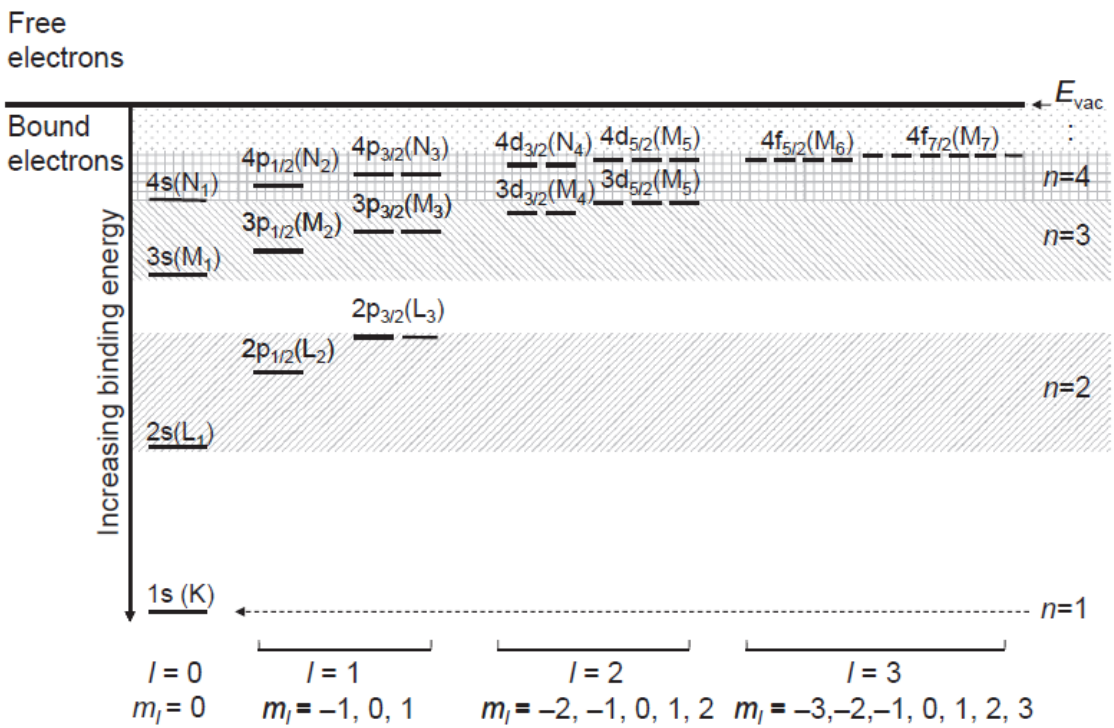




Electronic structure notion

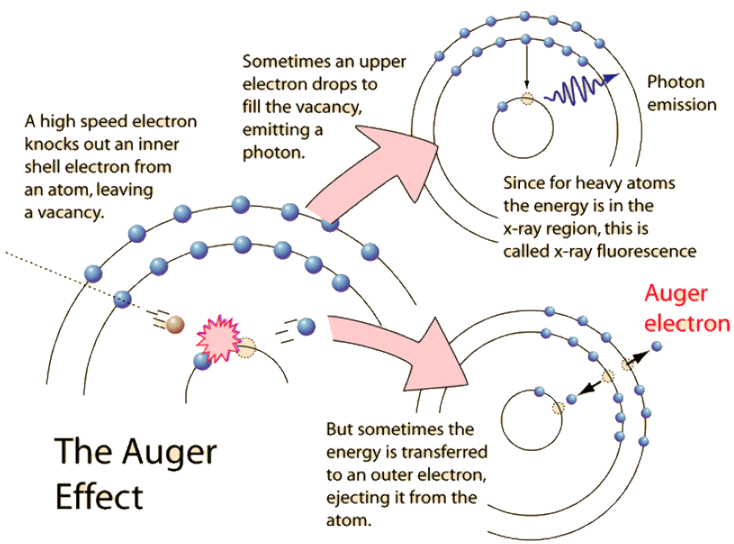
Spectroscopic Notation

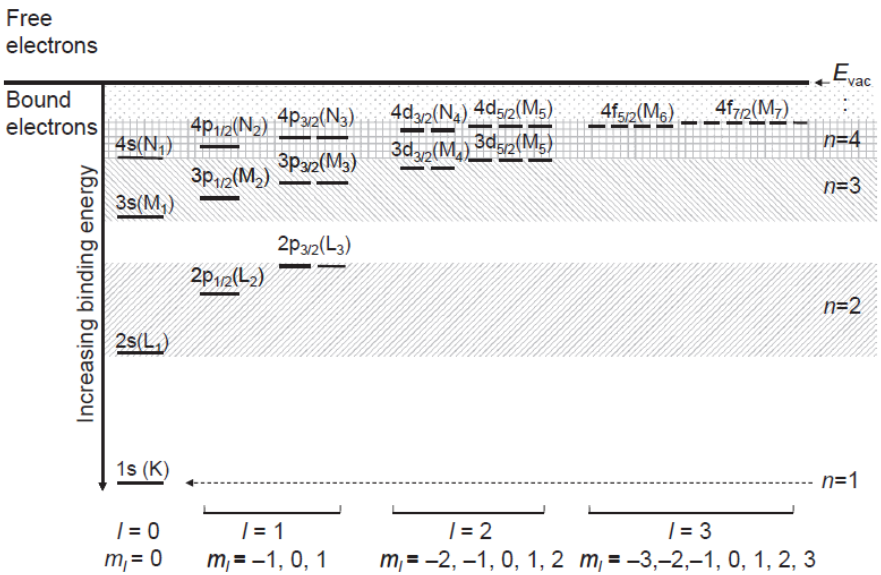
X-ray Notation



Electronic structure notion

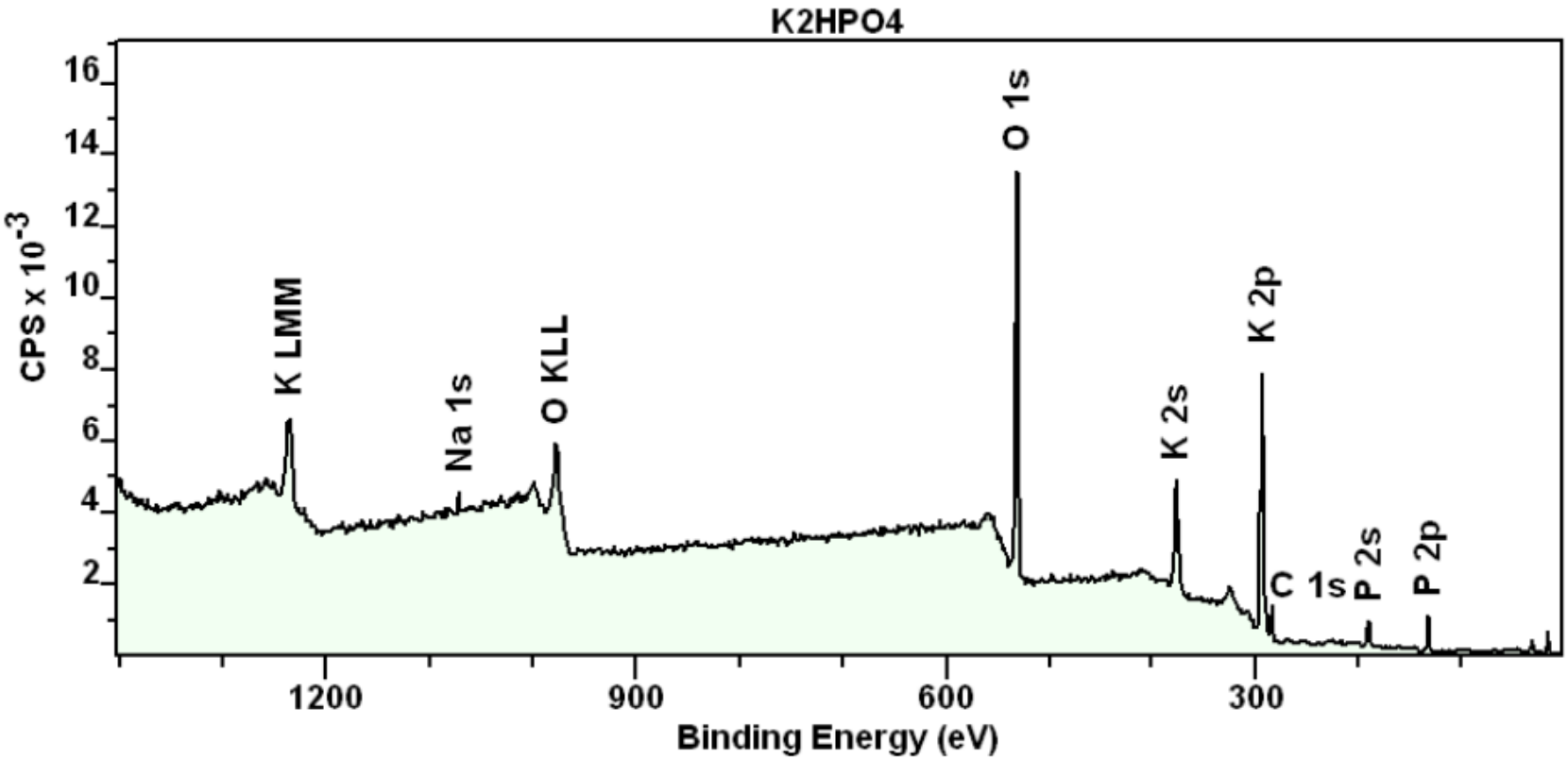
X-ray Notation





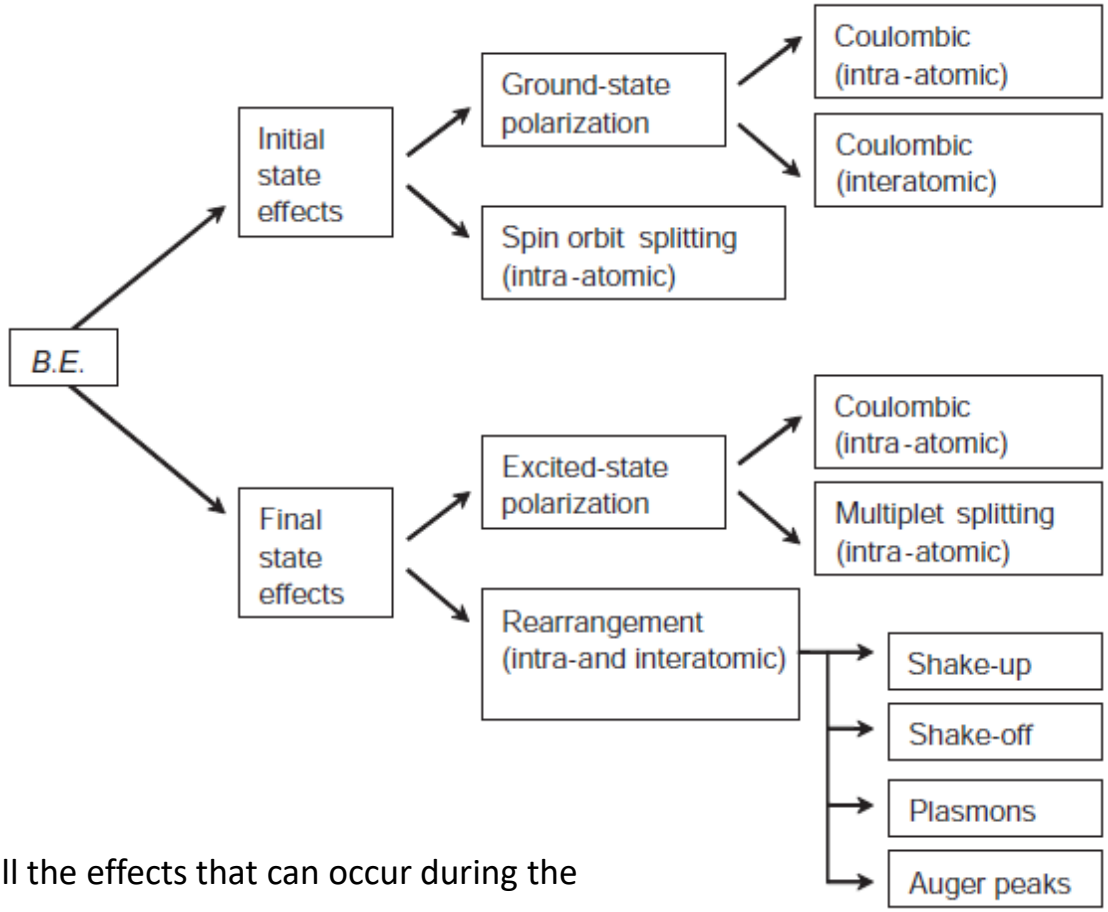
Electronic structure notion

X-ray Notation



Binding energy of photoelectron

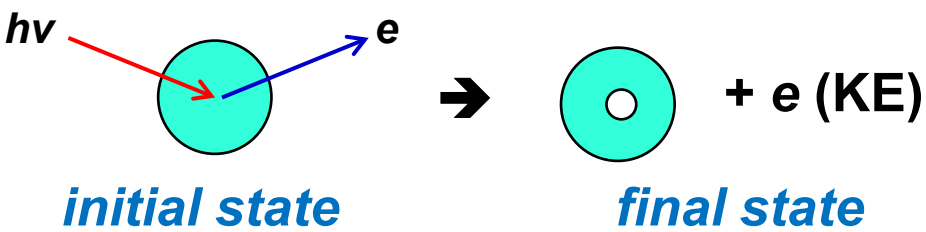
Speciation



"A schematic diagram depicting all the effects that can occur during the photoelectron emission process in systems bound to atoms/ions, in comparison to the binding energy (B.E.) values of unexcited free atoms."

Binding energy of photoelectron

$A(N) + h\nu \rightarrow A^{+}(N-1) + e$



$$E_i(N) + h\nu = E_f(N-1) + KE_e$$

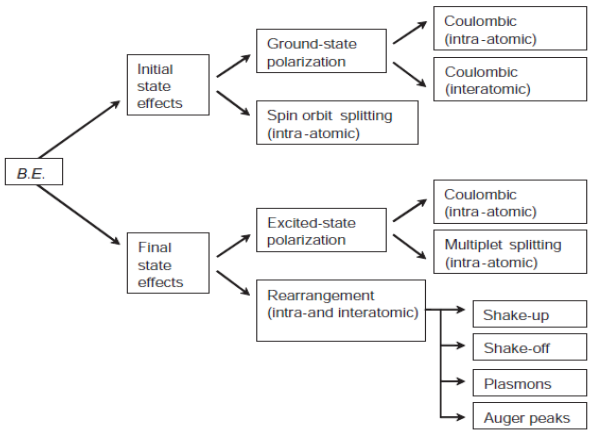
$$\mathrm{BE} = h\nu - KE_e = E_f(N-1) - E_i(N)$$

$$= -\varepsilon_i^{\mathrm{HF}} - \varepsilon_{relax} + \varepsilon_{correl}$$

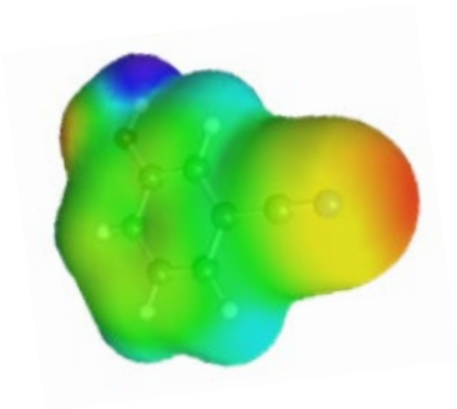
Koopmans' approximation

BE of core level ≈ $-\varepsilon_i^{\mathrm{HF}}$ (*i* th orbital energy)

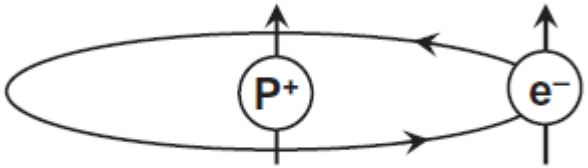
Binding energy of photoelectron: initial state



Coulombic interaction

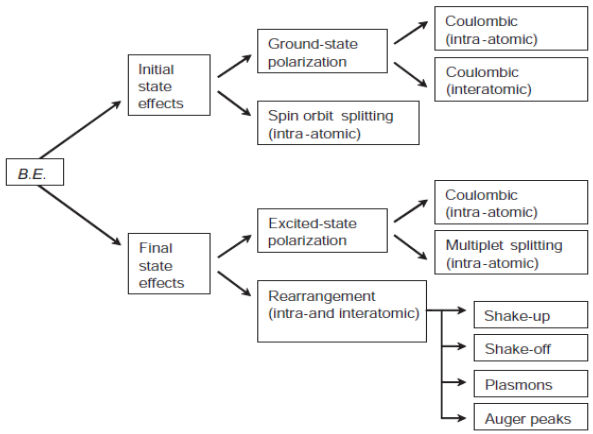


Spin-induced interactions



$j = l + s$

Binding energy of photoelectron: initial state



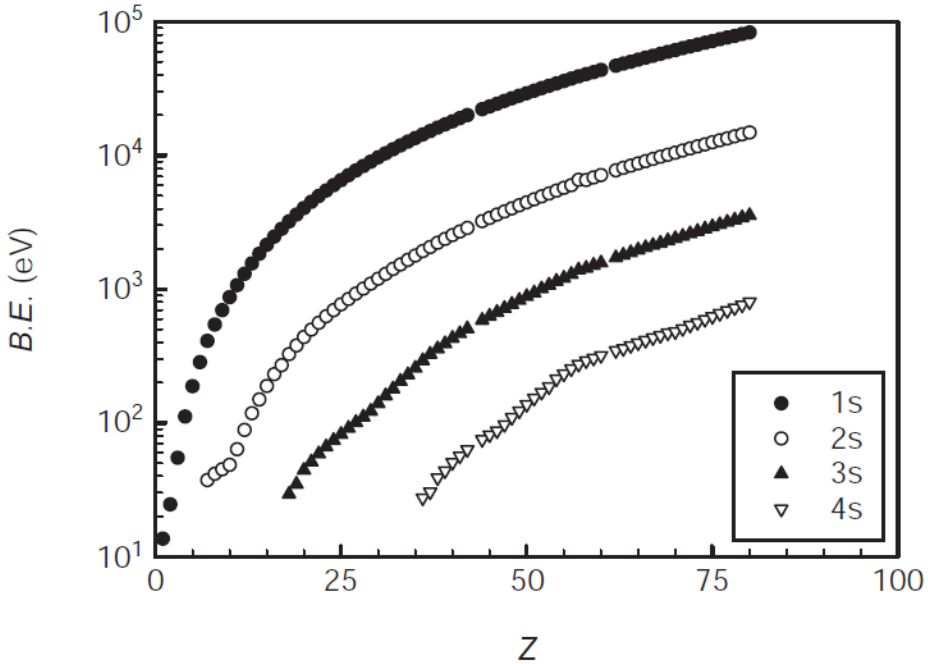
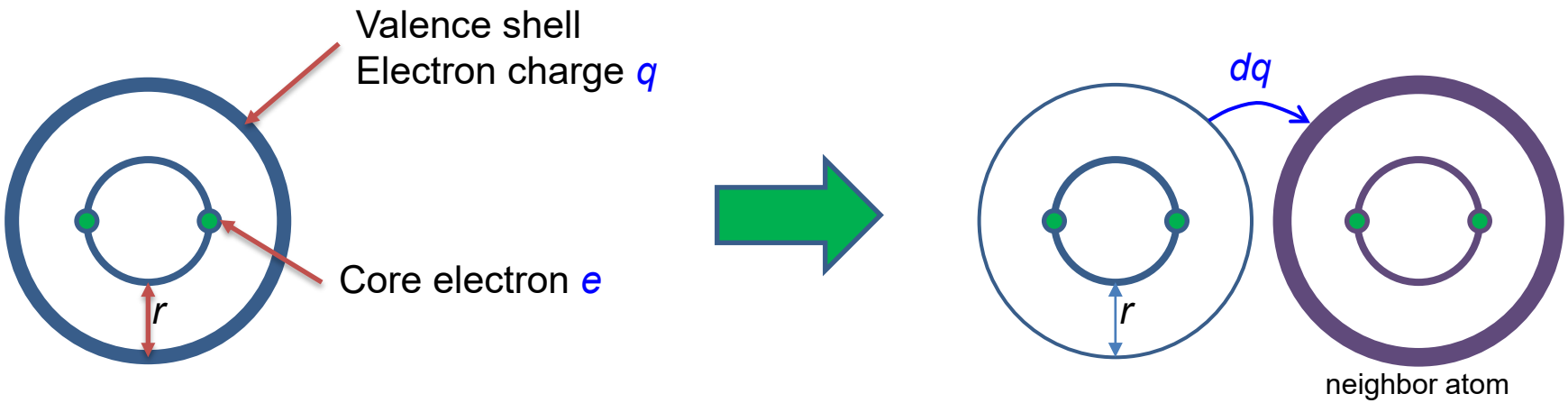


Figure 2.3. Binding energies of the 1s, 2s, 3s, and 4s electrons as a function of *Z* from all the naturally occurring elements from Li–Hg as measured by XPS and XRF (these values are listed in Appendix B).

Binding energy of photoelectron: initial state



The core electron (*e*) feels an alternation in the chemical environment when a change in the charge (*q*) of the valence shell occurs.

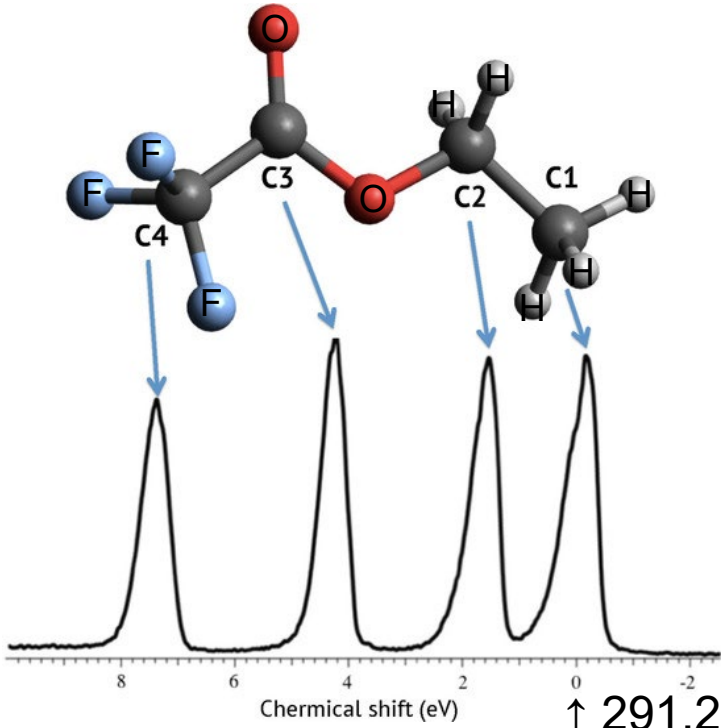
A change in *q*, *dq*, gives a potential change $dE = \frac{1}{4\pi\varepsilon_0}\frac{e(dq)}{r}$

- **The oxidation state of the atom**
- **The chemical environment**
  - electronegativity of neighbor atoms
  - number of neighbor atoms

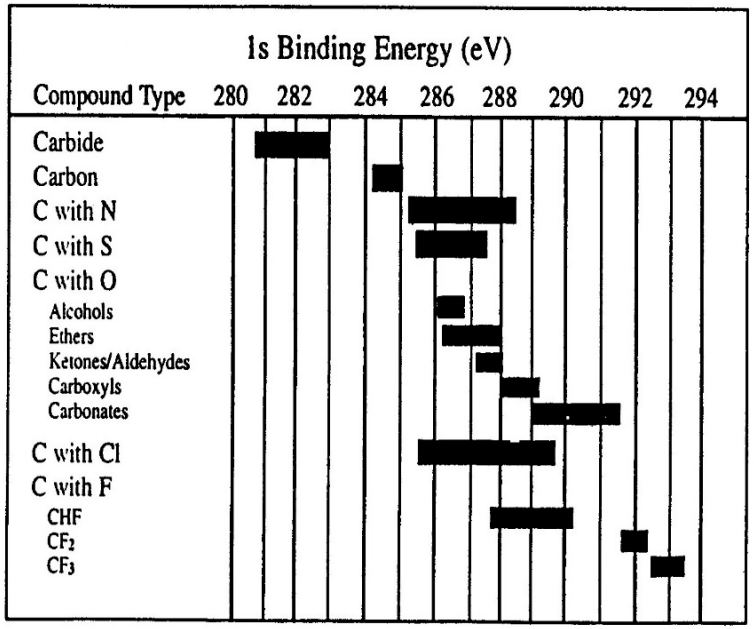
Binding energy of photoelectron: initial state

Not Always Possible

Tabulated Values Vary Widely



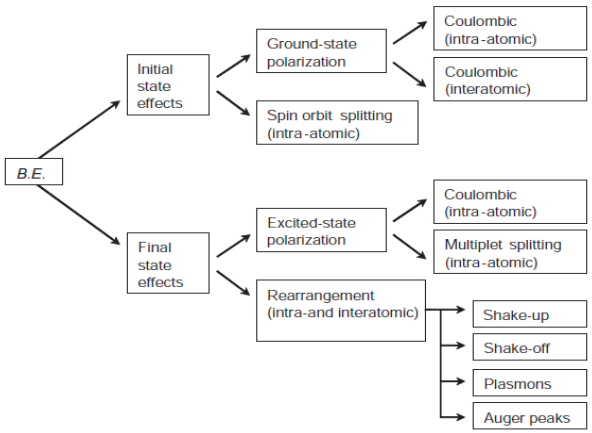
↑ 291.2 eV



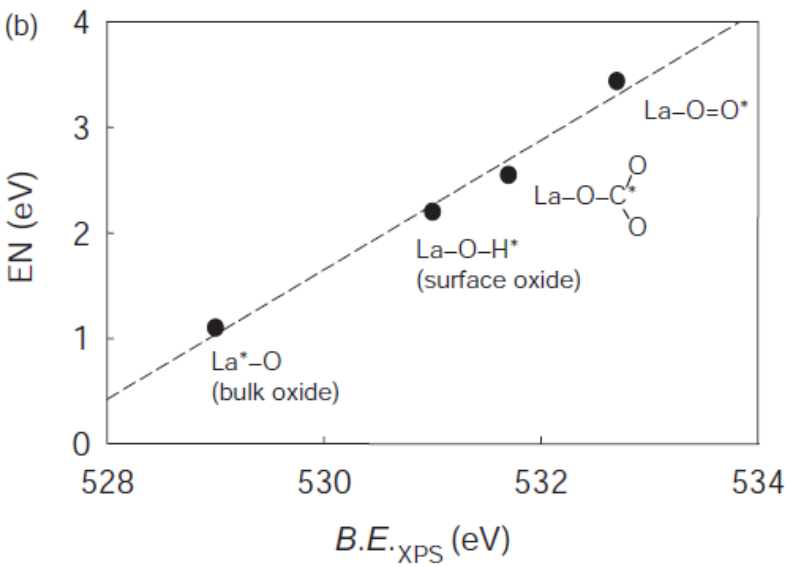
XPS handbook, PHI inc.

$$BE = E_F - E_i$$

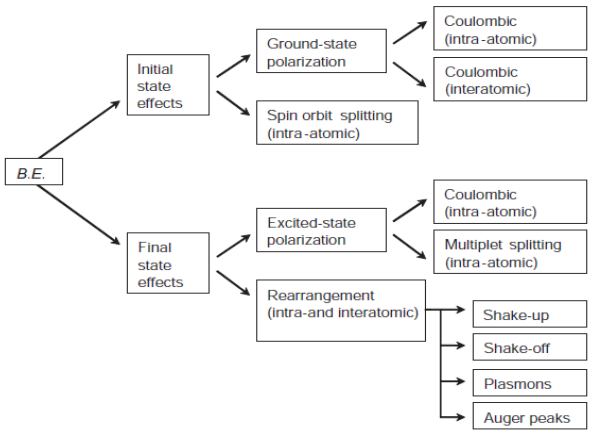
Binding energy of photoelectron: initial state



Coulombic interaction



Binding energy of photoelectron: initial state



Coulombic interaction

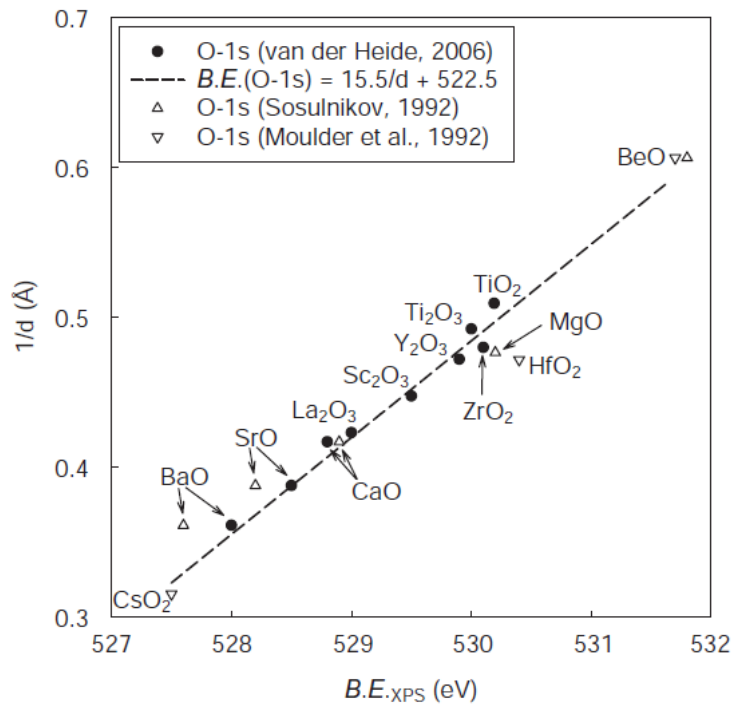
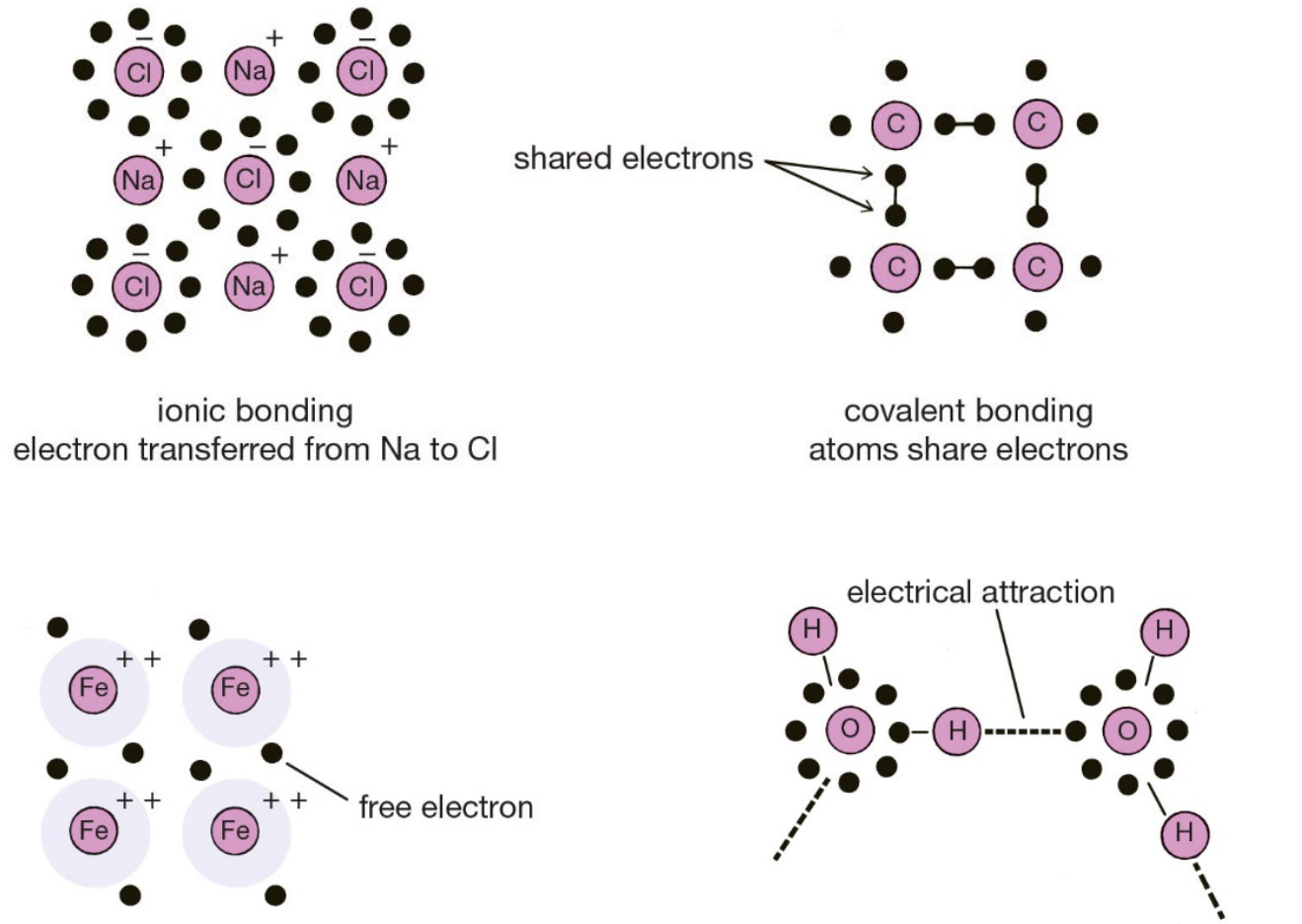


Figure 5.7. Photoelectron $B.E._{XPS}$ values as a function of inverse interatomic distance ($1/d$) for various single-component oxides. The dashed line represents the linear regression analysis of the data. Reproduced with permission from van der Heide (2006).

(a) They exhibit significant variations in $\Delta B.E.$ on oxidation; that is, beryllium exhibits an increasing $B.E.$XPS, whereas barium exhibits a decreasing $B.E.$XPS (values are listed in Table 5.3).
(b) The **metals exhibit the same oxidation state** variations (loss of two electrons).
(c) The oxides, with the exception of BeO, exhibit **similar structures.**
(d) The photoelectron emissions from these elements exhibit trends consistent with initial state effects (any final state effects are therefore either minimal or constant).

XPS spectrum interpretation

Speciation



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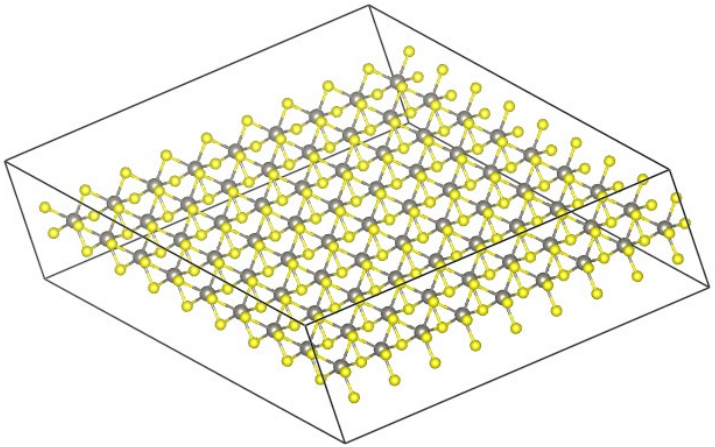
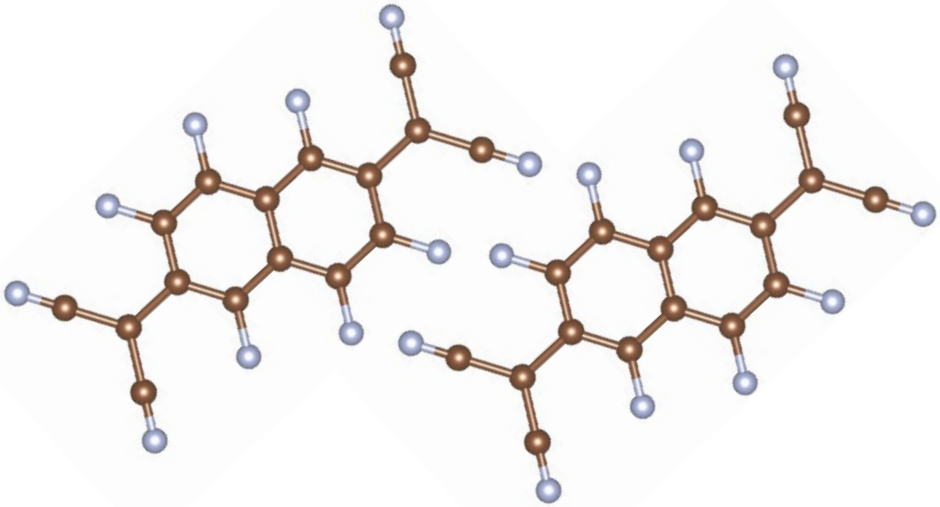
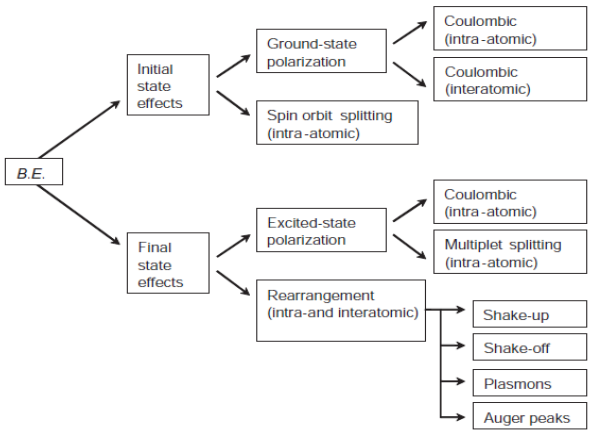
XPS spectrum interpretation

Speciation





Binding energy of photoelectron: initial state



Spin-induced interactions

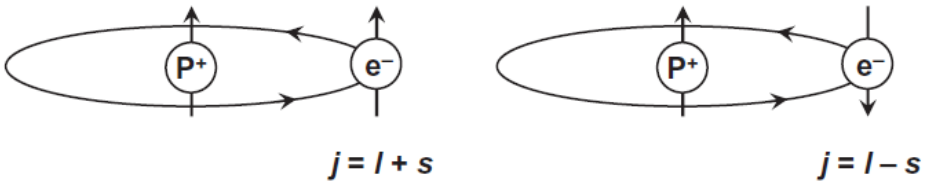


Figure 5.5. Schematic illustration of spin orbit coupling. The up arrows represent $m_s = 1/2$ and the down arrows $m_s = -1/2$. The orbit represents some nonzero orbit described by l.

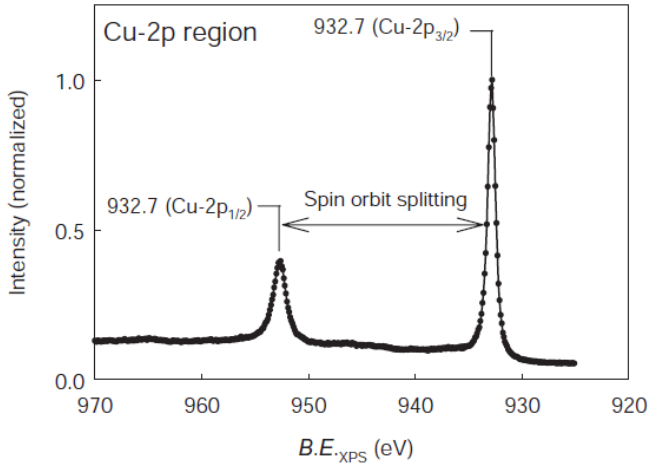
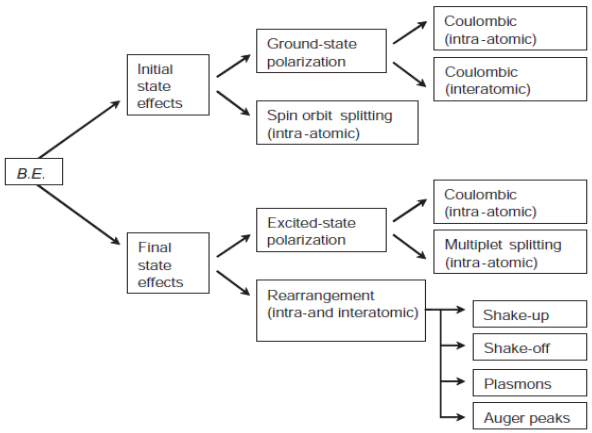


Figure 5.4. Spin orbit splitting exhibited by the Cu-2p photoelectrons from a copper metal surface. The data were collected using a concentric hemispherical analyzer (CHA) set at low-pass energy.

Binding energy of photoelectron: initial state



Spin-induced interactions

TABLE 5.2 Spin Orbit Splitting Factors for Electrons in the Specified Orbits (Specified Using Spectroscopic Notation), Their l and m_s Numbers, the Magnitude of J, and the Relative Peak Areas of the Spin Orbit Split Doublets

Orbital	l	m_s	J $(l + m_s)$	Area ratio $(2J + 1)$
p	1	+1/2, −1/2	3/2, 1/2	1:2
d	2	+1/2, −1/2	5/2, 3/2	2:3
f	3	+1/2, −1/2	7/2, 5/2	3:4

$$D_{SOS} = 2J + 1.$$

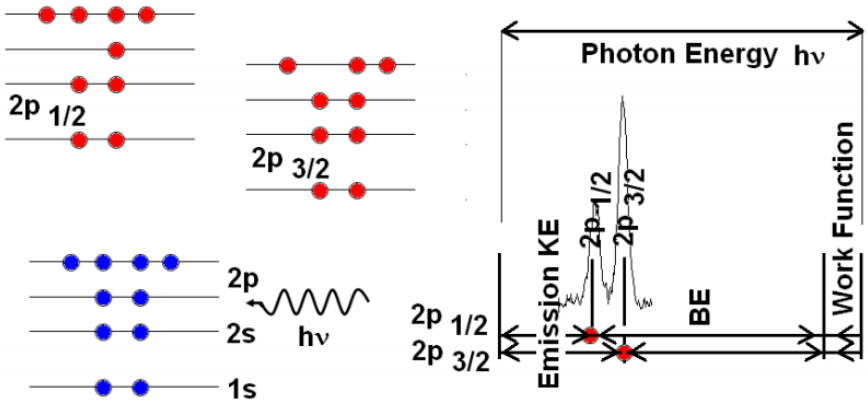
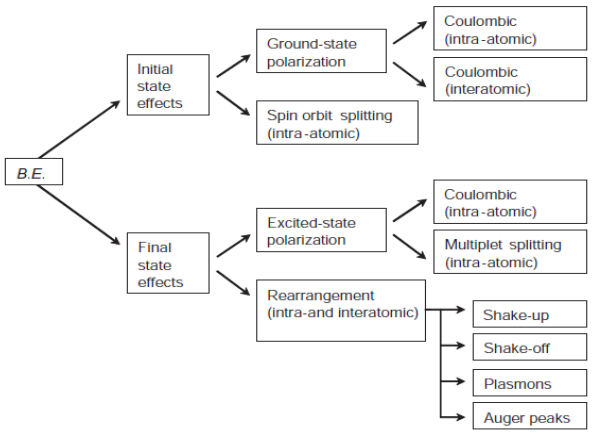
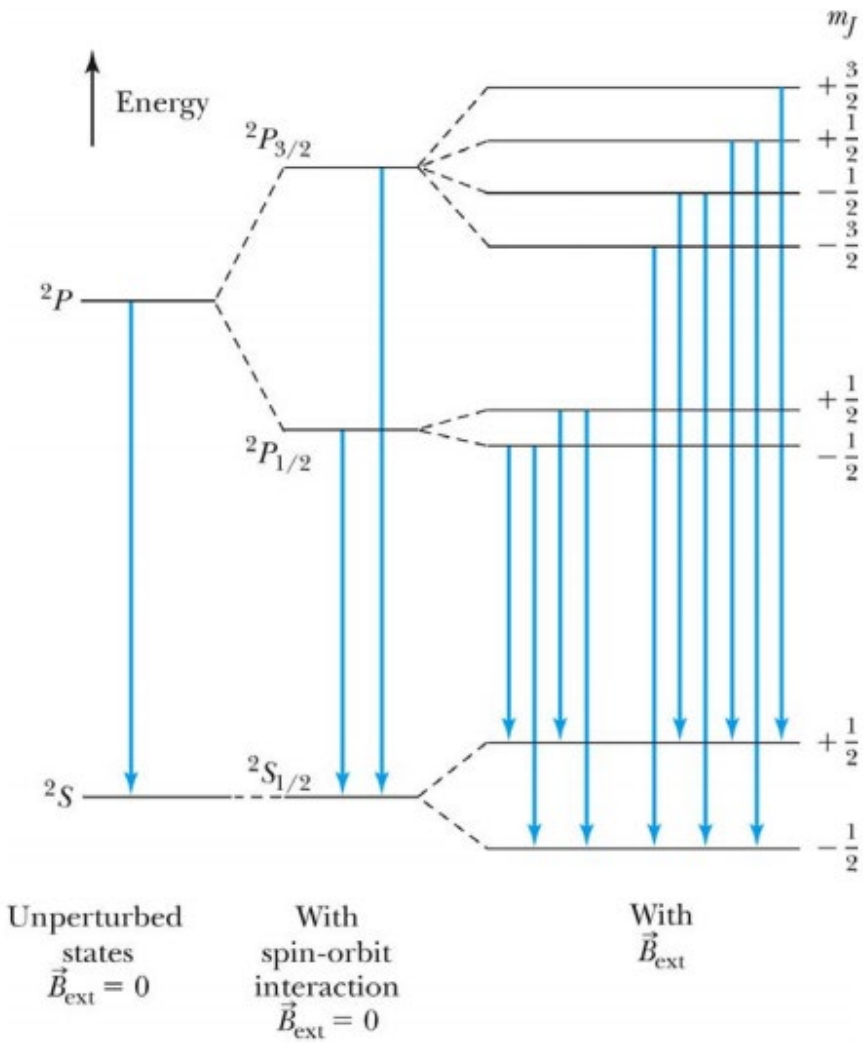


Figure 9: Excitation of L-shell electron resulting in doublet emission peaks.

Binding energy of photoelectron: initial state



Spin-induced interactions



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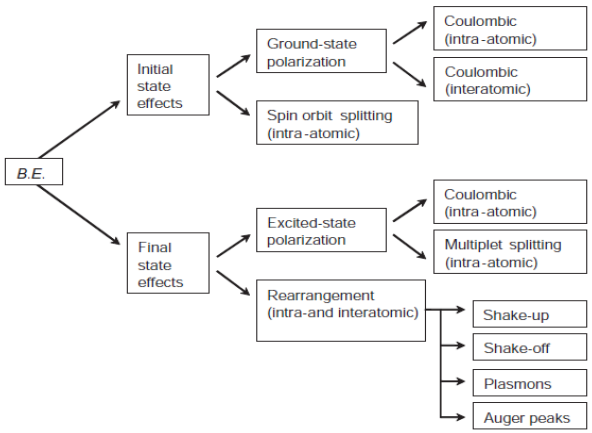
Binding energy of photoelectron: initial state



Spin-induced interactions

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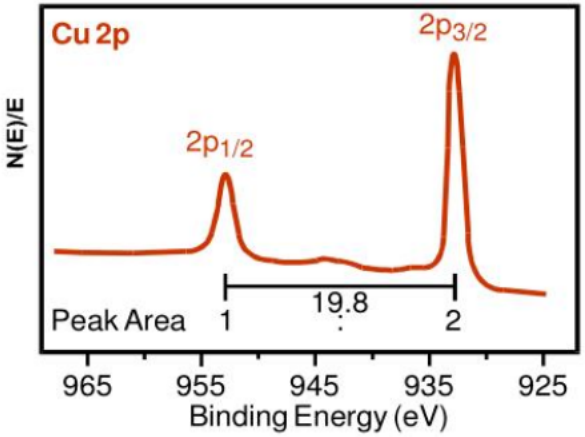
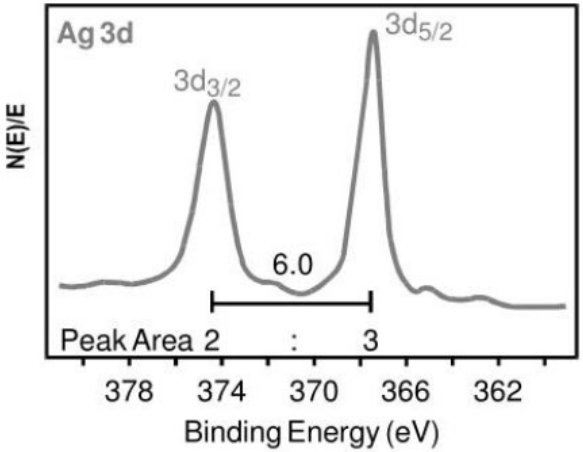
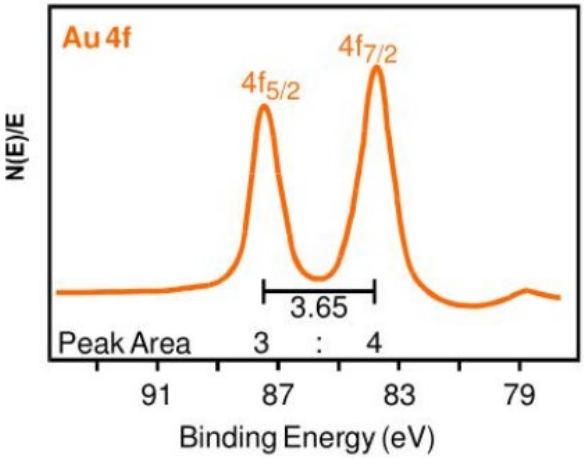
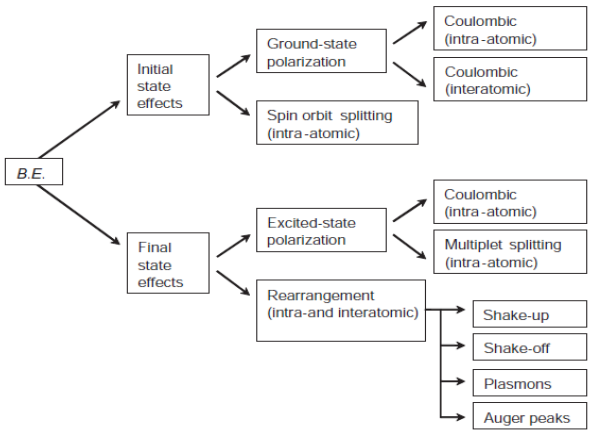






Binding energy of photoelectron: initial state



Spin-induced interactions

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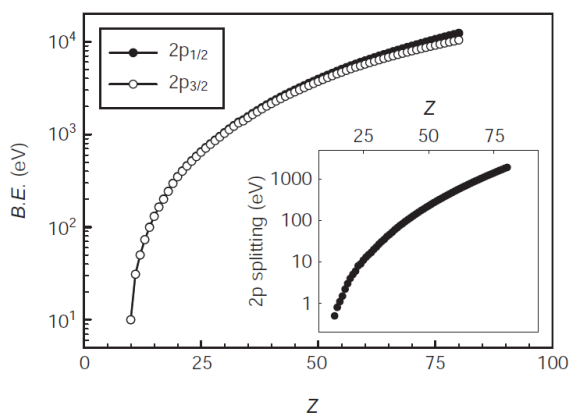
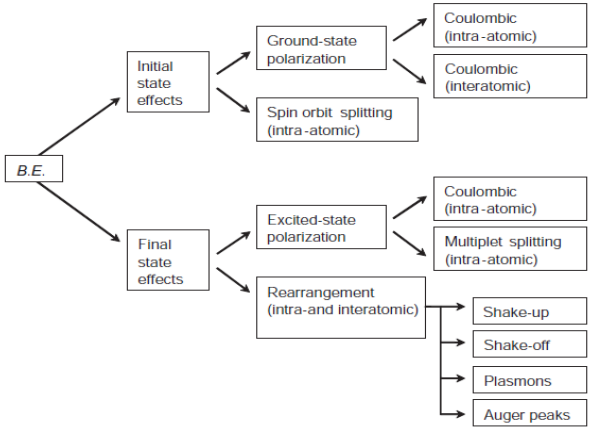


Figure 2.4. Binding energies of the $2p_{1/2}$ and $2p_{3/2}$ electrons as a function of Z for the naturally occurring elements from Li–Hg as measured by XPS and XRF (these values are listed in Appendix B). The difference between these levels, that is, the spin orbit splitting energy versus Z, is shown in the inset.

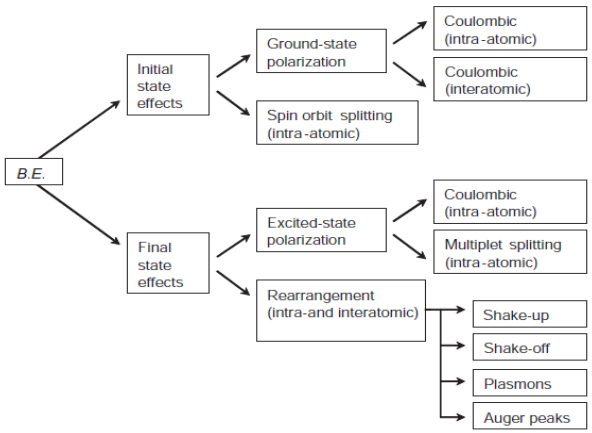
Binding energy of photoelectron: Final state



(a) Core hole-induced polarization
(instantaneous electrostatic and magnetic effects)

(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Binding energy of photoelectron: Final state



(a) Core hole-induced polarization
(instantaneous electrostatic and magnetic effects)

- Multiple splitting

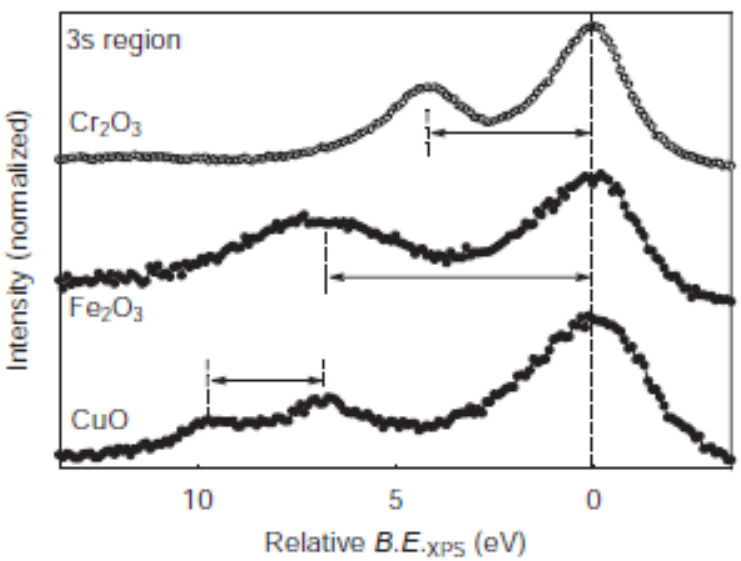
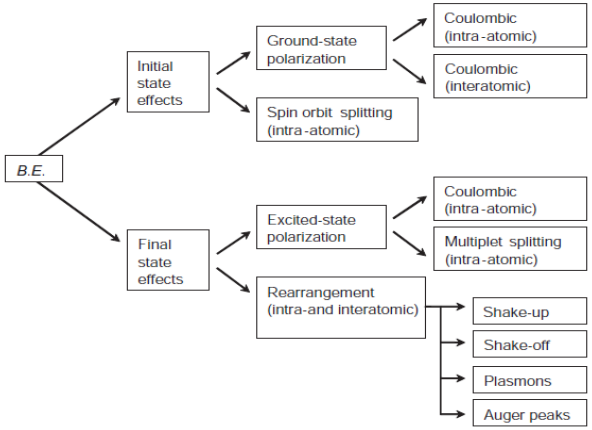


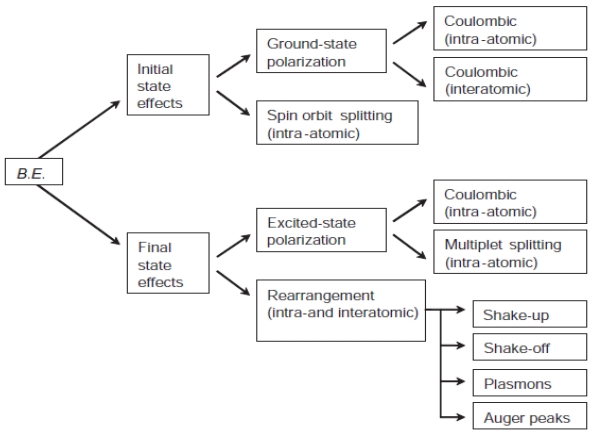
Figure 5.10. Multiplet splitting over the transition metal 3s regions from Cr_2O_3, Fe_2O_3, and CuO (CHA set to low-pass energy). The splitting is represented by the arrows. For comparative purposes, the main photoelectron $B.E._{XPS}$ is set to zero. *Note*: The splitting of the Cu-3s spectrum from CuO occurs on the 3s satellite, not the main peak as in Cr_2O_3 and Fe_2O_3.

Binding energy of photoelectron: Final state



(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Binding energy of photoelectron: Final state



(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Shake-up processes

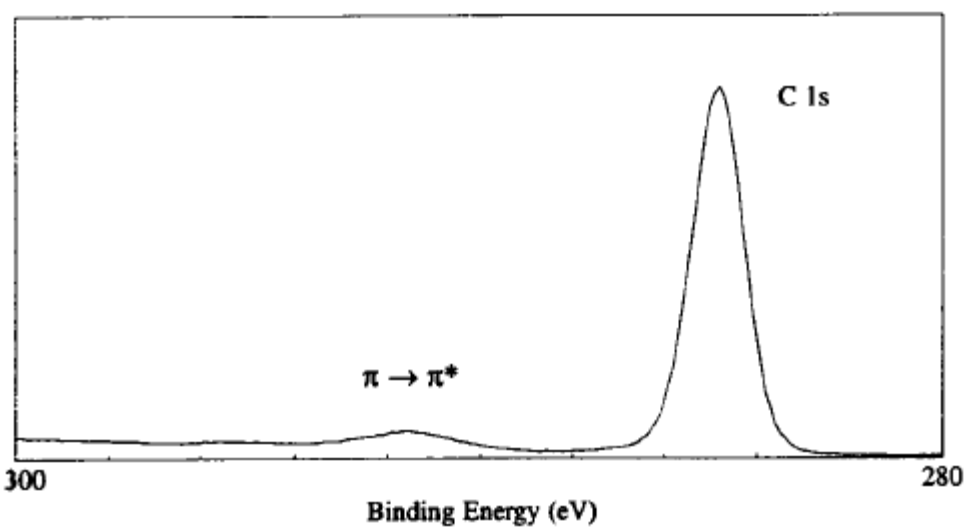
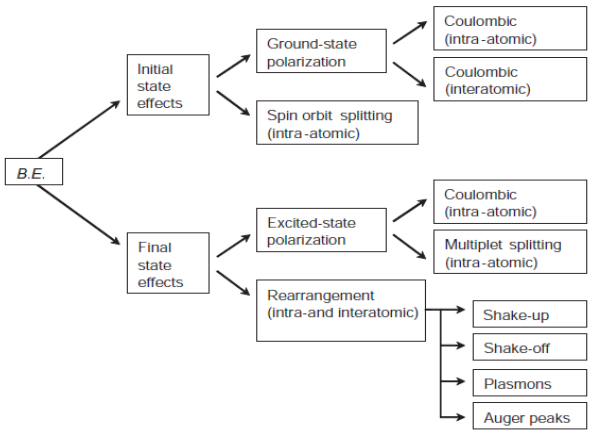


Figure 7. The π bond shake-up satellite for C 1s in polystyrene. The peak is about 6.7 eV higher than the main photopeak.

Binding energy of photoelectron: Final state



(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Shake-up processes

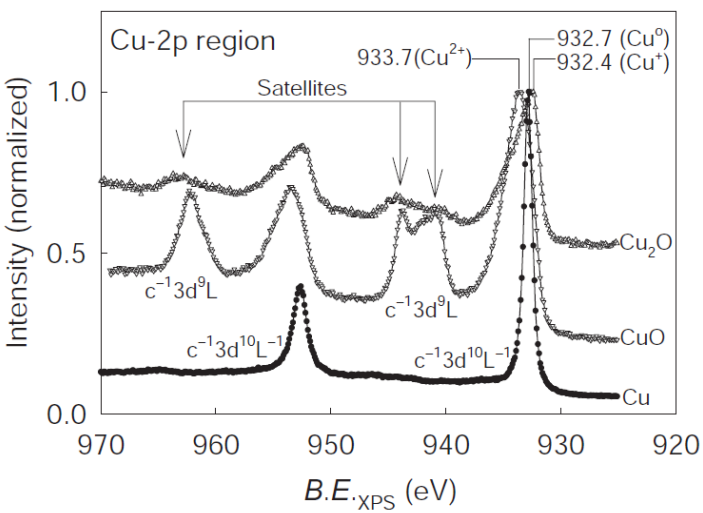
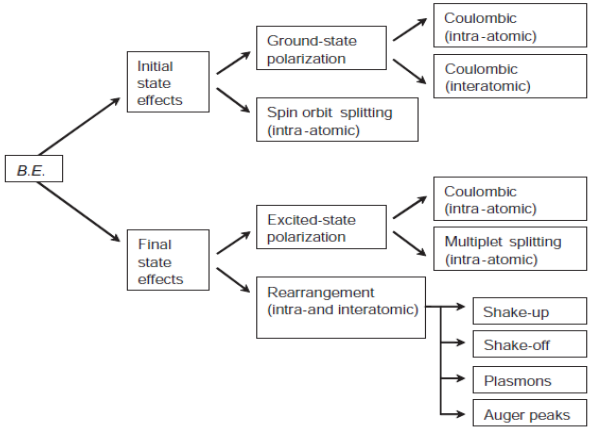


Figure 5.13. Photoelectron spectra over the Cu-2p region from the metal and oxides (Cu_2O and CuO) of copper collected under high-energy resolution conditions (CHA set to low-pass energy).

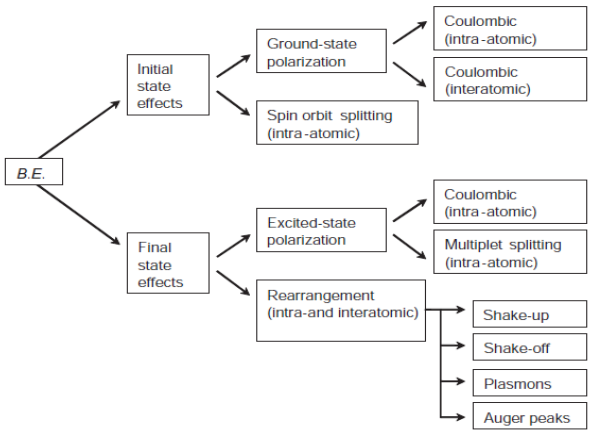
Binding energy of photoelectron: Final state



(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

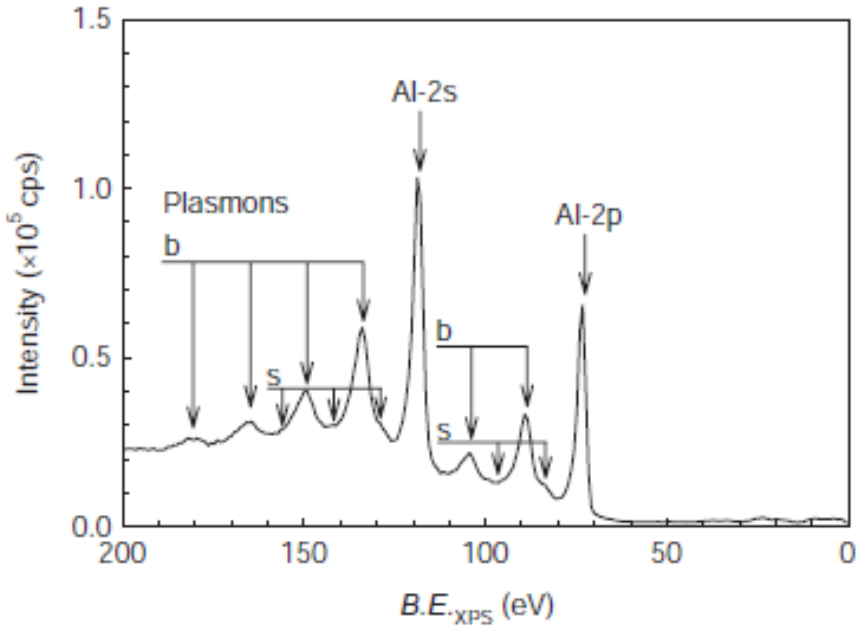
Shake-off processes

Binding energy of photoelectron: Final state

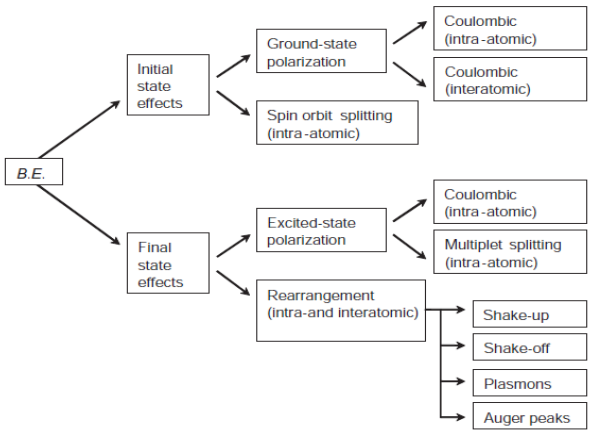


(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Plasmon Loss features

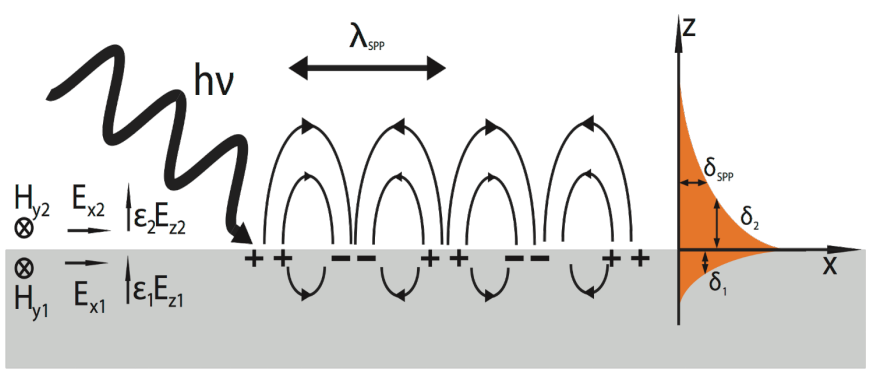


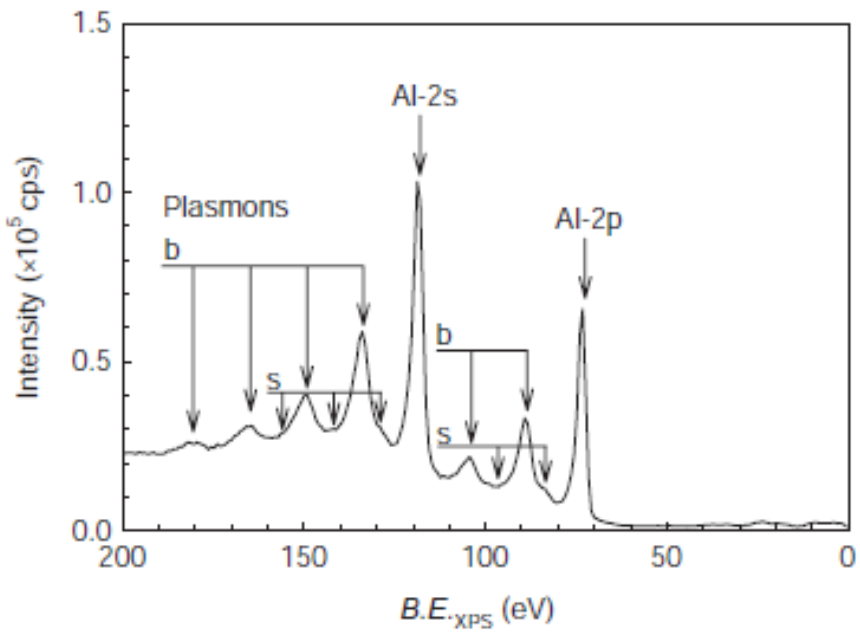
Binding energy of photoelectron: Final state



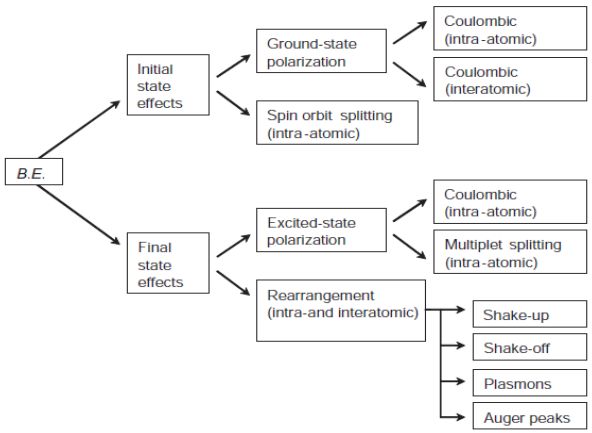
(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Plasmon Loss features



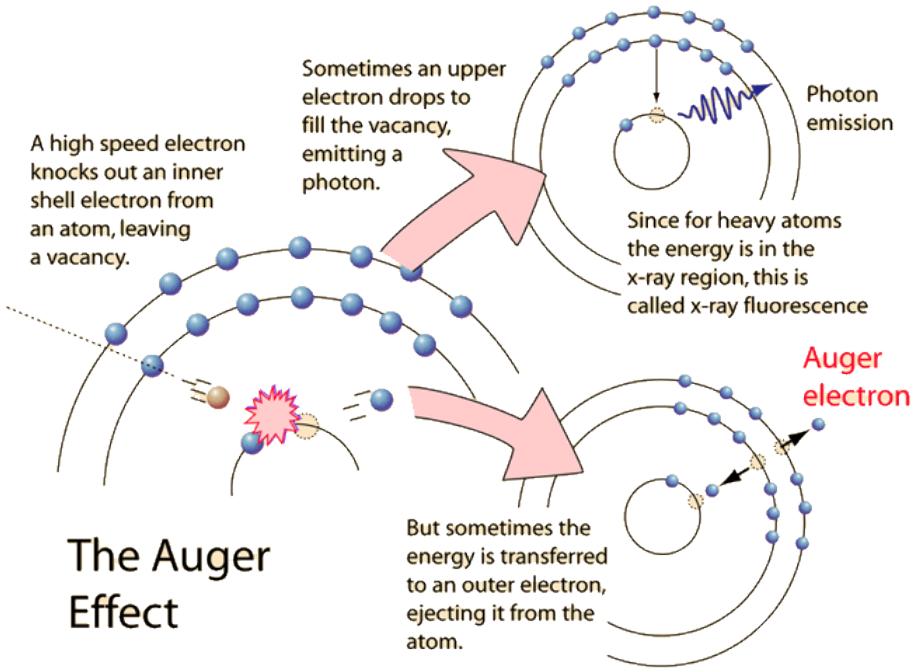


Binding energy of photoelectron: Final state



(b) Core hole-induced rearrangement
(subsequent excitation and relaxation processes)

Auger process



Thank you for listening